FALL 2026 ·

Welcome to CSE471

System Analysis and Design

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Hello, I'm Arif

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Research

Software Engineering, Software Testing, Program Analysis, Software Quality



Global collaborations

Research intern with teams at UIUC, George Mason, UT Austin, Auburn and UC Irvine.



Industry

Full-stack Trainee software engineer at Cefalo Bangladesh



Teaching at BRACU

Theory: CSE423 Computer Graphics, CSE470 Software Engineering, CSE471 SAD.

Labs: CSE423 and CSE370.

YOUR TURN

Introduce yourself in 30 seconds

01

Name

What would you like me to call you?

02

Outside class

A hobby, passion or skill you're proud of

QUICK CHECK-IN

Raise your hand if…



You've completed CSE470

Software Engineering — SDLC, MVC, design patterns, testing



You've built an app with a real database

ER models, SQL and a working project from CSE370



You've drawn a UML diagram before

Class, use case or sequence — any of them counts



A team project of yours went off track

Unclear requirements? Changing scope? We'll fix that here

No wrong answers — this tells me where to start.

THE BIG PICTURE

Where CSE471 fits in your journey

CSE370

Database

You learned to model and store data.

- ER / EER modeling
- Normalization
- SQL and a DB project

CSE470

Software Engineering

You learned how software gets built.

- SDLC and Agile
- MVC and design patterns
- Class diagrams, CFG and testing

YOU ARE HERE

CSE471

System Analysis and Design

Now: analyze and design a system before it is built.

- Feasibility and requirements
- Use case, activity and sequence diagrams
- Component, DFD and structure charts

THE BIG PICTURE

The SDLC, and where CSE471 lives in it

CSE471 FOCUS: understand and design the system

BUILD AND RUN: practised in CSE470 and CSE370

01
Planning
Is it worth building? Feasibility analysis

02
Analysis
What must it do? Requirements, use case, activity and sequence diagrams

03
Design
How will it work? Components, DFDs, structure charts, data storage and UI

04
Implementation
Write the code

05
Testing
Verify it works as specified

06
Deployment & Maintenance
Release, support and evolve the system

Why it matters: a mistake caught during analysis or design costs far less than one found after the code is written.

WHY IT MATTERS

Why analysis and design are worth learning

31%

of software projects were fully successful. The rest ran late, over budget or failed outright.

Standish Group, CHAOS Report 2020

47%

of unsuccessful projects missed their goals because of poor requirements management.

PMI, Pulse of the Profession 2014

$2.41T

estimated cost of poor software quality to the US economy in a single year.

CISQ, Cost of Poor Software Quality 2022

The pattern: many failures begin before any code is written, with unclear requirements and weak design. That is exactly what CSE471 trains you to get right.

THINK · PAIR · SHARE

Great systems are designed long before the first line of code.



Think of an app that frustrated you this week. Was it a coding problem — or a design problem?

Two minutes with your neighbour, then we share.

WHAT WE'LL EXPLORE

Our 10-week roadmap

Week 1	Introduction, SDLC, Waterfall and DevOps
Week 2	Feasibility analysis and requirements
Weeks 3–4	Behavioral UML: use case, activity, sequence
Week 5	**Midterm examination**
Weeks 6–7	Structural design: component and data flow diagrams
Week 8	Program design and structure charts
Week 9	Data storage design
Week 10	User interface design
Finals	**Final examination**
Throughout	4 assignments, 4 quizzes and weekly labs

Tentative plan — exact dates will be shared on the course page.

ASSESSMENT

How you'll be assessed

30

Lab

Weekly lab work

25

Midterm

Week 5

25

Final

End of semester

10

Quizzes

Best 3 of 4

10

Assignments

4 in total

Out of 100 marks



Labs carry the most weight — show up and engage every week.

LOGISTICS

When and where to find me

Class	Day	Time	Room
Theory · Sec 09	Sat & Thu	11:00 AM – 12:20 PM	10B-15C
Theory · Sec 10	Sat & Thu	12:30 – 1:50 PM	10B-15C
Lab · Sec 10	Tue	11:00 AM – 1:50 PM	10G-32L
Consultation	Sat	2:00 – 4:50 PM	4P176
Consultation	Thu	9:30–10:50 AM, 2:00–3:20 PM	4P176

Section 09 lab: as per your class routine.



Reach me

arif.hasan@bracu.ac.bd

Room 4P176

Tip: put your section and ID in the email subject.

WORKING TOGETHER

What we can expect from each other



From you

- Come on time, and come curious
- Ask questions: anytime, about anything
- Do your own work; academic integrity matters
- Reach out early if you are struggling



From me

- Clear explanations and materials
- Fair and transparent assessment
- Timely feedback on your work
- Open consultation hours — every question is welcome

BEFORE YOU GO

Let's design great systems together.

Exit question

In one word: what do you hope to take away from CSE471?

Questions?

arif.hasan@bracu.ac.bd · Desk 4P176

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